

Polynomials, factor and remainder theorems - Set 2

TMUA practice questions by topic. Specification MM1.6. 5 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Which of the following statements about the polynomial $p(x) = x^3 - 6x^2 + 11x - 6$ is/are true?
- I $p(x)$ has three distinct real roots.
 - II $(x - 4)$ is a factor of $p(x) - 6$.
 - III $p(x) > 0$ for all $x > 3$.
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III
- 2 The expression $x^4 + 4$ can be factorised into two quadratics with integer coefficients. What is the sum of the coefficients of x in these two quadratics?
- A 1
- B 0
- C 2
- D 4
- E -4
- F It cannot be factorised.
- 3 Given that $x^3 + px + q$ is divisible by $(x - 2)^2$, find the value of q .
- A 16
- B 12
- C 8
- D -8
- E -16
- F -12

- 4 When $x^3 - 3x^2 + 2x + 7$ is divided by $x^2 + 1$, what is the remainder?
- A $x + 4$
 - B $x + 10$
 - C 10
 - D $-x + 10$
 - E $x + 7$
 - F $3x + 4$
- 5 For which value of k does $x^2 - x + k$ divide exactly into $x^3 - 2x^2 + 3x - 4$? (Assume such a value exists.)
- A -2
 - B 3
 - C 4
 - D 2
 - E 1
 - F no such value exists